

Amplification of DNA from an individually isolated nucleus by use of the Xdrop™ system and method, - Version 2.

This protocol describes the lysis/denaturation of single isolated nucleus and generation of dMDA droplets including both DNA from a single nucleus and phi29 mastermix within the single emulsion droplets generated in the Xdrop microfluidic cartridge. The procedure includes the steps lysis of the nuclei and denaturation of its DNA, neutralization, mixing with dMDA reagents, injection into the dMDA cartridge, droplet formation, dMDA incubation, and isolation of amplified DNA.

The general Xdrop Instrument description, handling and loading of the dMDA cartridge as well as the enzyme mix recipes can be found in the [Xdrop manual](#).

This updated version introduces new steps to ensure better lysis and denaturation of the DNA inside the nuclei. This will be done by adding a heat denaturation step after the lysis / neutralization step. Also please look at the droplet formation in a microscope and evaluate the droplets formed. If we have issues with the quality of the droplets, we could also evaluate to reduce salt or rely only on heating step (benchmark them with or without lysis step).

NOTE: When working with single cells or nuclei, there is of course only two copies of each chromosome, and all possible measures should be taken to avoid losing any material. Obviously, it's important to minimize handling of lysed material before amplification has occurred. It's recommended to minimize excess mixing of volumes and using low bind plastic tubes and pipette tips.

Materials and Methods

Lysis of single isolated nucleus

Make a diluted lysis buffer by mixing 2.8 µL Lysis buffer (200 mM KOH, 5 mM EDTA (pH 8) and 40 mM 1.4 DTT) diluted with 2 µL PCR grade water in a low bind (if possible) 0.2 mL PCR vial.

One nucleus is added to per PCR vial with lysis buffer and lysed and DNA denatured here for 5 minutes at room temperature. Then 1.4 µL neutralization buffer (400 mM HCl and 600 mM Tris HCl (pH 7.5)) is added and incubated for 5 min at room temperature.

After chemical lysis is done we will add 5x dMDA buffer (Samplix dMDA kit item# RE20300) and water up to 19 μ L reaction volume and heat the sample in a thermal cycler at 95°C for 3 minutes and then let the samples cool on the bench for 5-10 minutes and then put them on ice.

dMDA master mix	1x
dMDA mix (5x)	4 μ l
H ₂ O	8.8 μ l
Lysed/neutralized nuclei*	6.2 μ l
Total volume	19 μ l
Heat denaturation	95 °C for 3 minutes
Cool at RT on the bench – spin down	5-10 minutes
Add dMDA enzyme	1 μ l
Total volume	20 μ l

Table 1: dMDA reagents, mastermix and volumes.

**We assume that it should be ok to dilute the lysis buffer in water. We think this is a better solution than having more volume of such high salt buffers will interfere with the droplet formation. This lysis plus heat denaturation should be sufficient for a complete lysis and access of the hexamers.*

Finally, when ready to load the dMDA cartridge add the 1 μ L dMDA enzyme and mix carefully (table 1) and spin shortly. Then, using a wide bore pipette, the 20 μ L MDA amplification mixture is injected into the Inlet site. While injecting the mixture into the Inlet site be carefull not to spill the mastermix, hold the pipette for 15 seconds before releasing. Then add 75 μ L dMDA oil (Samplix dMDA kit item# RE20300) into the overall inlet well (supply cavity).

MDA droplets generation.

The dMDA cartridge is closed with a gasket and moved into the Xdrop™ instrument and the single emulsion droplets are formed which typically takes ~ 45 s. Droplets are collected into low bind 0.2 ml PCR vials from the Collection container of the dMDA cartridge. Make sure to harvest as much droplets as possible by taking extra oil from the bottom of the 0.2 mL collection PCR vial and wash the collection container. After all droplets are collected, excess oil is removed from the collection PCR vials leaving only 2-3 mm of dMDA oil at the bottom of each vial.



MDA in droplets.

Move the dMDAs droplets in the 0.2 mL collection PCR vials to a thermal block and incubate the MDA droplets at 30° C for 16 hours and then heat inactivated at 65°C for 10 minutes. If using the thermal block with a hot lid make sure not to keep the lid hotter than 70-75° C during the incubation.

Breaking of droplets.

Droplets are broken by adding 20 µL Break solution and 1 µL Break colour (Samplix dMDA kit item# RE20300) (Samplix ApS, Herlev, Denmark) mix moderately by flicking (not vortex). Spin down briefly and remove the oil phase from the bottom but keep the aqueous phase (pink/reddish). Repeat spinning to fully remove the oil from the aqueous phase if needed.

DNA QC of Xdrop™ droplet MDA and bulk MDA.

DNA material from Xdrop™ droplet MDA and bulk MDA reactions may be quantified using Qubit™ Fluorometer (ThermoFisher Inc., Waltham, MA, USA) and the DNA integrity investigated using Fragment Analyzer (Agilent Inc., Santa Clara, CA, USA) according to the manufacturer's instructions.

Next Generation Sequencing.

It is recommended to make DNA libraries using TruSeq PCR free DNA library kit (Cat. No. 20015962, Illumina Inc., San Diego, CA, USA) or similar methodologies (limiting PCR cycling of the library if possible) after standard mechanical fragmentation (for example Covaris, Woburn, MA, USA) or enzymatic fragmentation (for example NEBNext® Ultra™ II FS DNA Module from New England Biolabs (cat. No. E7810S)).

Oxford Nanopore libraries can also be made directly from dMDA derived DNA using Native ligation / barcoding protocols from Oxford Nanopore with T7 Endonuclease I debranching step.



Appendix I:

Extra reagents needed (must be prepared in your own lab):

Lysis buffer: 200mM KOH + 5mM EDTA + 40mM DTT

- Store in aliquots at -80 degrees for no longer than three months.

Neutralization buffer: 400mM HCl + 600mM Tris-HCl, pH 7.5

- Store in aliquots at -80 degrees.



Appendix II:

QC of droplet productions. As mentioned, the salt in the buffers may cause the dMDA droplets to be unstable. It is therefore a good idea to look at the droplets in a microscope to be sure of the droplet structure and formation.

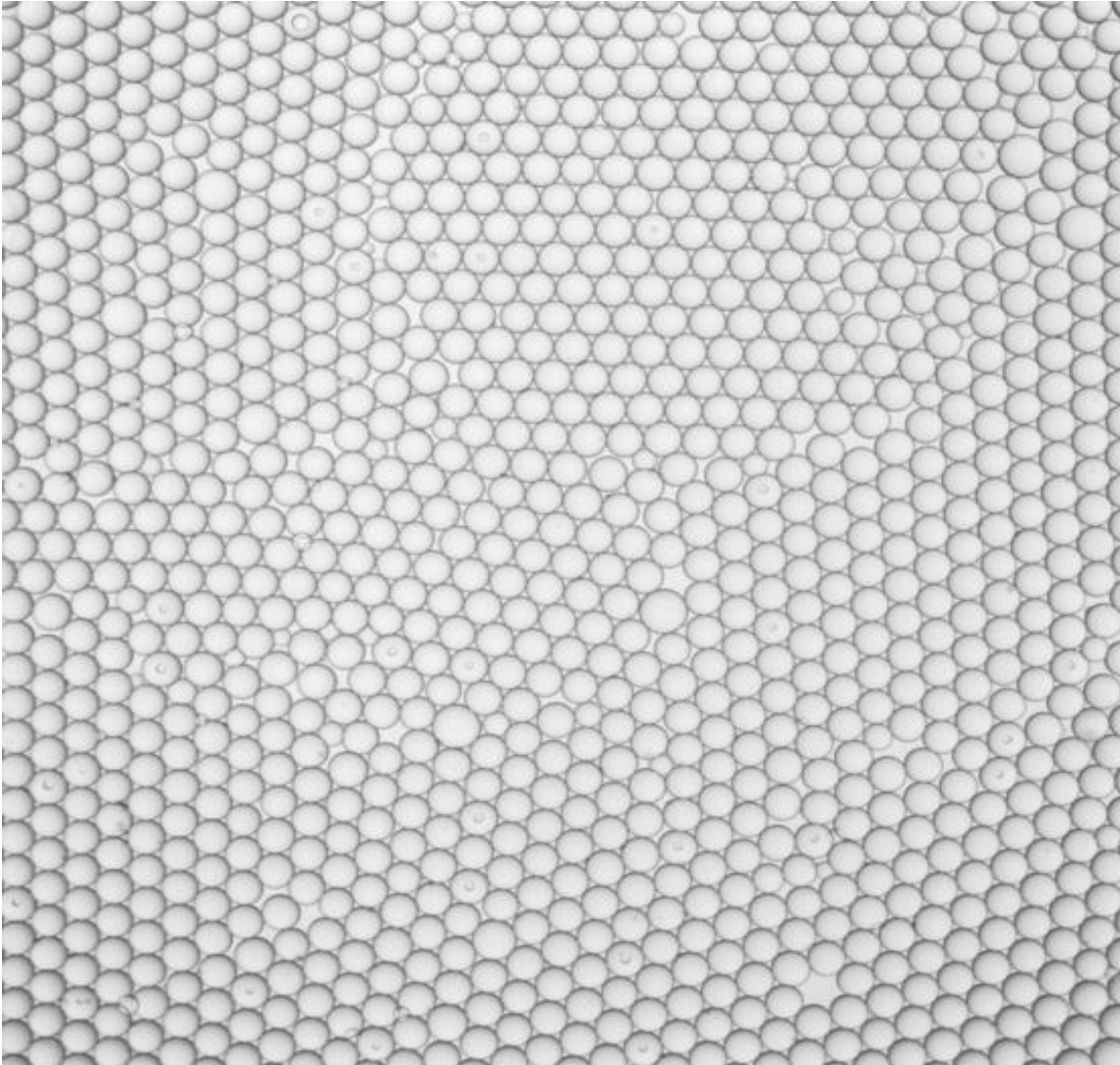


Figure 1: Microscope dMDA droplets after droplet generation (brightfield x4). These droplets look perfect. The droplet we get out of a lysis /neutralization single cell amplification reaction will probably never look as good as these but this is the “**golden standard**”.

